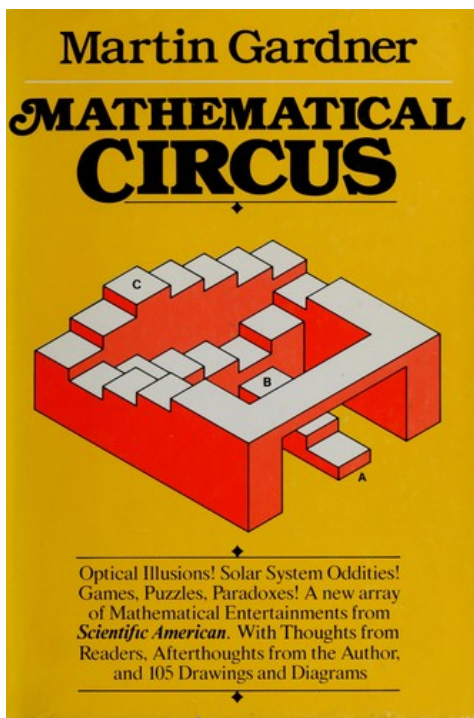


Puzzles to Puzzle You (Remastered)

Welcome to the twelfth iteration of *Puzzles to Puzzle You (Remastered)* by Rani Rachavelpula, in which I will discuss the previous puzzle (Bugs on a Polyhedron) and pose the next one.

This will be the final issue of the newsletter. I can't believe we made it through a full year of puzzles. Thank you for puzzling along, thinking hard, and making it all worthwhile. At present, there are no plans to continue into 2026. I trust we all possess the strength to cope with the absence. To assist with the transition, however, I am recommending the book *Mathematical Circus* from the beloved Martin Gardner.

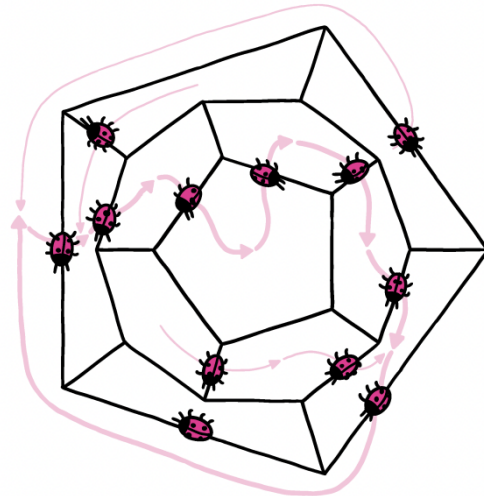


Mandatory Reading

Puzzle #11 Bugs on a Polyhedron

Associated with each face of a solid convex polyhedron is a bug which crawls along the perimeter of the face, at varying speed, but only in the clockwise direction.

1. Prove that no schedule will permit all the bugs to circumnavigate their faces and return to their initial positions without incurring a collision.



Bugs on a Regular Dodecahedron

Shout Out (Rishi) December is for family, so the final shout out of the year goes to my brother Rishi. In the Vedas, a Rishi is a sage—an exalted, divine human being who earns eternal knowledge through intense meditation (*tapas*). I would not, as a rule, describe my brother in these terms. However, this past month, our conversations about puzzles and Terence Tao have come close enough to justifying his namesake.

Solution To solve this puzzle, we can think of the polyhedron as a system which evolves over time as the bugs move. Our task is to prove that the system must reach a certain state—one of collision—before all bugs complete a circuit. Thus, our goal is to identify some parameter that measures progress towards the desired state. Such a parameter is called a *potential*.

As is often the case, identifying the right potential involves a lot of trial and error. In this case, the potential is a little bit complicated and requires some detailed construction.

Without loss of generality, assume that each bug begins on some edge (as opposed to a vertex). Recall, each face F_i has a corresponding bug b_i , where b_i is located on an edge bordering two faces. For a fixed F_i , draw an arrow from the center of F_i through b_i to the center of the bordering face F_j , as depicted in the above

image. Iterate this step. Locate the bug b_j corresponding to F_j , where this bug now lies on an edge bordering faces F_j and F_k . Now, draw an arrow from the center of F_j through b_j to the center of F_k . Continuing in this way, we will eventually hit one face a second time, completing a cycle of arrows on the polyhedron.

What we have done is construct a loop on the surface of the polyhedron (Think of a rubber band on the surface on the polyhedron). By the Classical Jordan Curve Theorem, our loop divides the surface of the polyhedron into two parts — an ‘inside’ and an ‘outside.’ Let the inside be defined as the region to the right as one is walking along the loop. Let P be the number of vertices of the polyhedron contained in the inside of the loop. This P is our potential.

Initially, P can be anything from zero to all of the polyhedron’s vertices. Notice that P is 0 if and only if two bugs are on the same edge of the polyhedron. Since these bugs travel in opposite directions along the shared edge, they are destined to collide.

Because the bugs only move clockwise, the value of P can only decrease. As a bug passes a vertex, like the hands of a clock, its corresponding arrow rotates clockwise and that vertex moves from the inside of the loop to the outside. Further notice that there is no way for P to increase. Thus, each time a bug crosses a vertex, P decreases by 1. To complete a cycle, each vertex must be traversed at least three times, so P must monotonically decrease to 0. This implies that two bugs will necessarily collide. $\square$

Prelude to the Final Puzzle

For the final puzzle, I want something that each of you can keep in your back pocket anytime you want to impress at a dinner party or gala. This is a puzzle that does not require pen and paper and comes with a little mathematical lore. It can be militantly approached with sharp mental computation or it can be effortlessly solved if one sees the trick. You have the choice of hammer or scalpel, and both, I would say, are satisfying and impressive.

Puzzle #12 Two Bicyclists and a Fly

Two bicyclists start at opposite ends of a road 20 miles long. Each cyclist travels toward the other at 10 miles per hour. When they begin, a fly sitting on the front wheel of one of the bikes takes off and races at 15 miles per hour toward the other bike. As soon as it gets there, it instantly turns around and zips back toward the first bike, then back to the second, and so on. It keeps flying back and forth until it’s finally squished between their front tires when the bikes collide.

1. How far did the fly travel, in total, before it was squished?



Solution One way to solve the problem is to write out the distance the fly travels as a geometric series and compute its sum. A simpler way, however, is to focus on the cyclists instead of the fly. Given that they are 20 miles apart and each travels 10mph, we know that the cyclists will take exactly 1 hour to collide. Thus, the fly is flying for a total of 1 hour at 15mph. Hence, the fly will have traveled 15 miles.

In a now famous scene from John Nash’s *A Beautiful Mind*, the inimitable polymath John von Neumann (with whom I share a birthday) solved the problem before a computer. When von Neumann heard the puzzle, he instantly replied, ‘15 miles.’ His disappointed questioner said, “Oh, you saw the trick.” “What trick?” said von Neumann. “I just summed the infinite series.”



John von Neumann ♡